

YASH DIWAN

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EDUCATION

SHALOM HILLS INTERNATIONAL SCHOOL

Class 12 Percentage: **92.6%**

May 2023

Class 10 Percentage: **90.3%**

May 2021

B.Tech in Computer Science and Engineering, VIT Bhopal University

2023 - 2027

Relevant coursework: Data Structures and Algorithms, Operating Systems, Database Management Systems, and Software Engineering | **CGPA: 8.66/10.00**

SKILLS

Programming Languages: Java, Python

Developer Tools: VS Code, Jupyter Notebook, Google Colab, Git, GitHub

Technologies & Frameworks: Flask, FastAPI, Scikit-learn, NumPy, Pandas, Matplotlib, Streamlit

Databases: MySQL, MongoDB

ML Concepts: Supervised Learning, Feature Engineering, Data Analysis, EDA

PROJECTS

ImpACT ([Github](#))

Python | Node.js | MongoDB | YOLOv8s | React

Built a Pan-India multimodal web platform that automates civic grievance triage and municipal officer routing.

Engineered a custom YOLOv8s computer vision model over 100 epochs with advanced augmentations to detect physical hazards, achieving an **mAP@50 of 0.865 and 0.89 precision** on real-world citizen images.

Architected a Python/Flask ML microservice decoupled from the Node.js backend to serve real-time visual inferences and geospatial routing, sustaining <450ms latency under 500 concurrent requests.

Heal-Bridge AI ([Github](#))

Python | Flask | Scikit-learn | Pickle | HTML/CSS/JS

Engineered a health prediction web app for real-time disease diagnosis based on symptoms.

Engineered a robust 3-tier web application using **Flask** and **Machine Learning** pipelines to deliver real-time disease diagnosis, achieving **85%+ prediction accuracy** across a dataset of 100+ symptoms.

Revamped the user input interface by implementing a categorized selection system, cutting symptom entry time by **60%** and significantly enhancing the user experience.

Next.js | FastAPI | Google Maps API | Scikit-learn | Python | WebSockets

Simhashta ([Github](#))

Safety control room for the world's largest human gathering – Simhashta Kumbh Mela.

Engineered a real-time routing engine utilizing A* search algorithm to autonomously reroute pilgrims away from stampede-risk zones, **achieving a 10.85ms pathfinding latency** across a 5,000-node graph.

Architected a high-concurrency FastAPI backend with WebSockets to broadcast live crowd fluctuations at 2-second intervals, integrating a Gradient Boosting model to forecast 39.3M+ visitors based on 40 years of data.

CERTIFICATIONS

IBM – IBM Generative AI Engineering

([Links](#))

ORACLE – Oracle Database@AWS Certified Architect Professional

NPTEL – Introduction to Machine Learning

IBM – Machine Learning with Python with Honors

Google – Bits and Bytes of Computer Networking